

JEE MAIN REVISION MASTER: MATHEMATICAL REASONING

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Mathematical Reasoning — Core Principles & Governing Laws	8 PYQs	Core Principle
Standard	Mathematical Reasoning — Standard Calculations & Problem Archetypes	2 PYQs	Quantitative Analysis
Standard	Mathematical Reasoning — Applied Cases, Reactions & Traps	0 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Mathematical Reasoning — Core Principles & Governing Laws	$P(n):2^n < n!$, $\quad 3^n$, $\quad x;x^2+5$
Mathematical Reasoning — Standard Calculations & Problem Archetypes	3^n , $\quad P(n):2^{2n}-1$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Mathematical Reasoning — Core Principles & Governing Laws (High Yield — 8 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Mathematical Reasoning.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Mathematical Reasoning.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Mathematical Reasoning PYQs.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The solution of $3(x-1) \leq 2(x-3)$ is..." [Ans: A]
- [2024 Shift] Q#2 (MCQ): "The negation of the statement "For every real number $x;x^2+5$ is positive" is..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "The contrapositive of the statement. "If two lines do not intersect in the same plane, then they are parallel." is..." [Ans: N/A]

3.2 Mathematical Reasoning — Standard Calculations & Problem Archetypes (Standard Yield — 2 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Mathematical Reasoning.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2018 Shift] Q#3 (MCQ): " $P(n): 2^{2n-1}$ is divisible by k for all $n \in \mathbb{N}$ " is true, then the value of ' k ' is..." [Ans: N/A]
- [2018 Shift] Q#4 (MCQ): "The negation of the statement " 72 is divisible by 2 and 3 " is..." [Ans: N/A]

3.3 Mathematical Reasoning — Applied Cases, Reactions & Traps (Standard Yield — 0 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Mathematical Reasoning.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The solution of $3(x-1) \leq 2(x-3)$ is..." [Ans: A]
- [2024 Shift] Q#2 (MCQ): "The negation of the statement "For every real number x ; x^2+5 is positive" is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The solution of $3(x-1) \leq 2(x-3)$ is

- (A) $x \leq -3$ [Correct]
- (B) $x \geq -3$
- (C) $x \leq 3$
- (D) $x \geq 3$

Correct Answer: A

Solution Analysis: Given inequality: $3(x-1) \leq 2(x-3)$ Open the brackets: $3x-3 \leq 2x-6$ Now subtract $2x$ from both sides: $x-3 \leq -6$ Add 3 to both sides: $x \leq -3$ So, the solution is: $x \leq -3$ Hence, the correct option is Option A .

Benchmark Problem 2: KCET 2018 (Q#3)

$P(n): 2^{2n-1}$ is divisible by k for all $n \in \mathbb{N}$ is true, then the value of ' k ' is

- (A) 6
- (B) 3
- (C) 7
- (D) 2

Correct Answer: None

Solution Analysis: To determine the value of k such that $P(n) = 2^{2n-1}$ is divisible by k for all $n \in \mathbb{N}$, we start by examining a specific case: Consider $n=1$: $P(1) = 2^2 \times 1 - 1 = 2^2 - 1 = 4 - 1 = 3$ Since $P(1) = 3$, it follows that $P(n)$ is divisible by 3. Therefore, the value of k is 3.